



EAST WEST UNIVERSITY

Assignment -03

Course Details

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Network Topology

The network consists of three routers connected through serial DCE. Each router is also connected to its own local area network (LAN) via a switch, where two PCs are connected. Each router uses a different routing protocol like OSPF, RIP and EIGRP. The topology diagram provided below illustrates the complete setup, including all interfaces and assigned IP addresses.

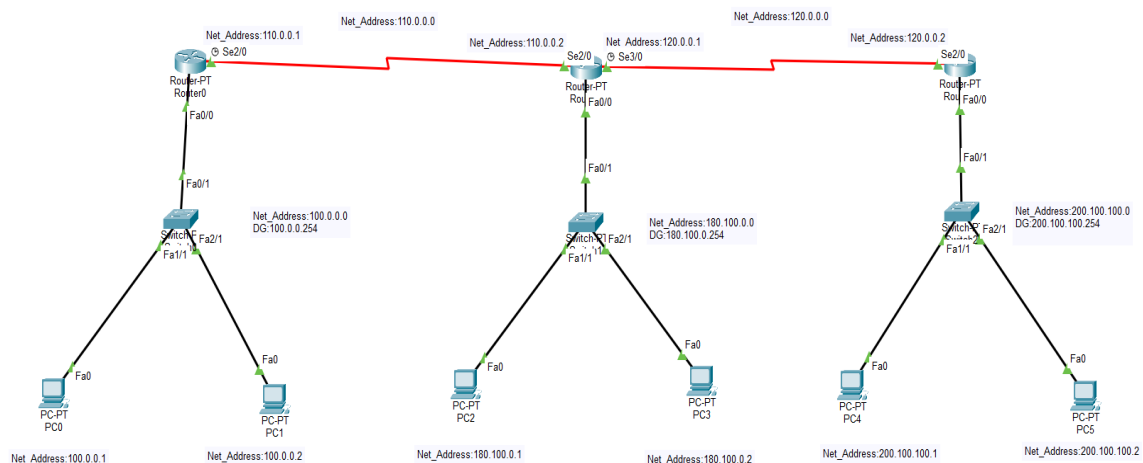


Figure 1: Network diagram with OSPF, RIP and EIGRP Router Protocol

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Network	Default Gateway
Router0	Fa0/0	100.0.0.254	255.0.0.0	100.0.0.0	—
	Se2/0	110.0.0.1	255.0.0.0	110.0.0.0	—
Router1	Fa0/0	180.100.0.254	255.255.0.0	180.100.0.0	—
	Se2/0	110.0.0.2	255.0.0.0	110.0.0.0	—
	Se3/0	120.0.0.1	255.255.0.0	120.0.0.0	—
Router2	Fa0/0	200.100.0.254	255.255.255.0	200.100.0.0	—
	Se2/0	120.0.0.2	255.255.0.0	120.0.0.0	—
PC0	Fa0	100.0.0.1	255.0.0.0	100.0.0.0	100.0.0.254
PC1	Fa0	100.0.0.2	255.0.0.0	100.0.0.0	100.0.0.254
PC2	Fa0	180.100.0.1	255.255.0.0	180.100.0.0	180.100.254

PC3	Fa0	180.100.0.2	255.255.0.0	180.100.0.0	180.100.0.254
PC4	Fa0	200.100.100.1	255.255.255.0	200.100.100.0	200.100.100.254
PC5	Fa0	200.100.100.2	255.255.255.0	200.100.100.0	200.100.100.254

Router Configurations

For Router 0:

fa0/0

```
Router>en
Router#config t
Router(config)#int fa0/0
Router(config-if)#ip address 100.0.0.254 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

se2/0

```
Router(config)#int se2/0
Router(config-if)#ip address 110.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 100.0.0.254 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip address 110.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
```

Router0 (RIP Configuration)

```
Router(config)#router rip
Router(config-router)#network 110.0.0.0
Router(config-router)#network 100.0.0.0
Router(config-router)#exit
```

```
Router(config)#router rip
Router(config-router)#network 110.0.0.0
Router(config-router)#network 100.0.0.0
Router(config-router)#exit
```

Router0 (OSPF Configuration)

```
Router(config)#router ospf 1
Router(config-router)#network 110.0.0.0 0.255.255.255 area 1
Router(config-router)#network 100.0.0.0 0.255.255.255 area 1
Router(config-router)#exit
```

```
Router(config)#router ospf 1
Router(config-router)#network 110.0.0.0 0.255.255.255 area 1
Router(config-router)#network 100.0.0.0 0.255.255.255 area 1
Router(config-router)#exit
```

Router0 (EIGRP Configuration)

```
Router(config)#router eigrp 1
Router(config-router)#network 110.0.0.0 0.255.255.255
Router(config-router)#network 100.0.0.0 0.255.255.255
Router(config-router)#exit
```

```
Router(config)#router eigrp 1
Router(config-router)#network 110.0.0.0 0.255.255.255
Router(config-router)#network 100.0.0.0 0.255.255.255
Router(config-router)#exit
```

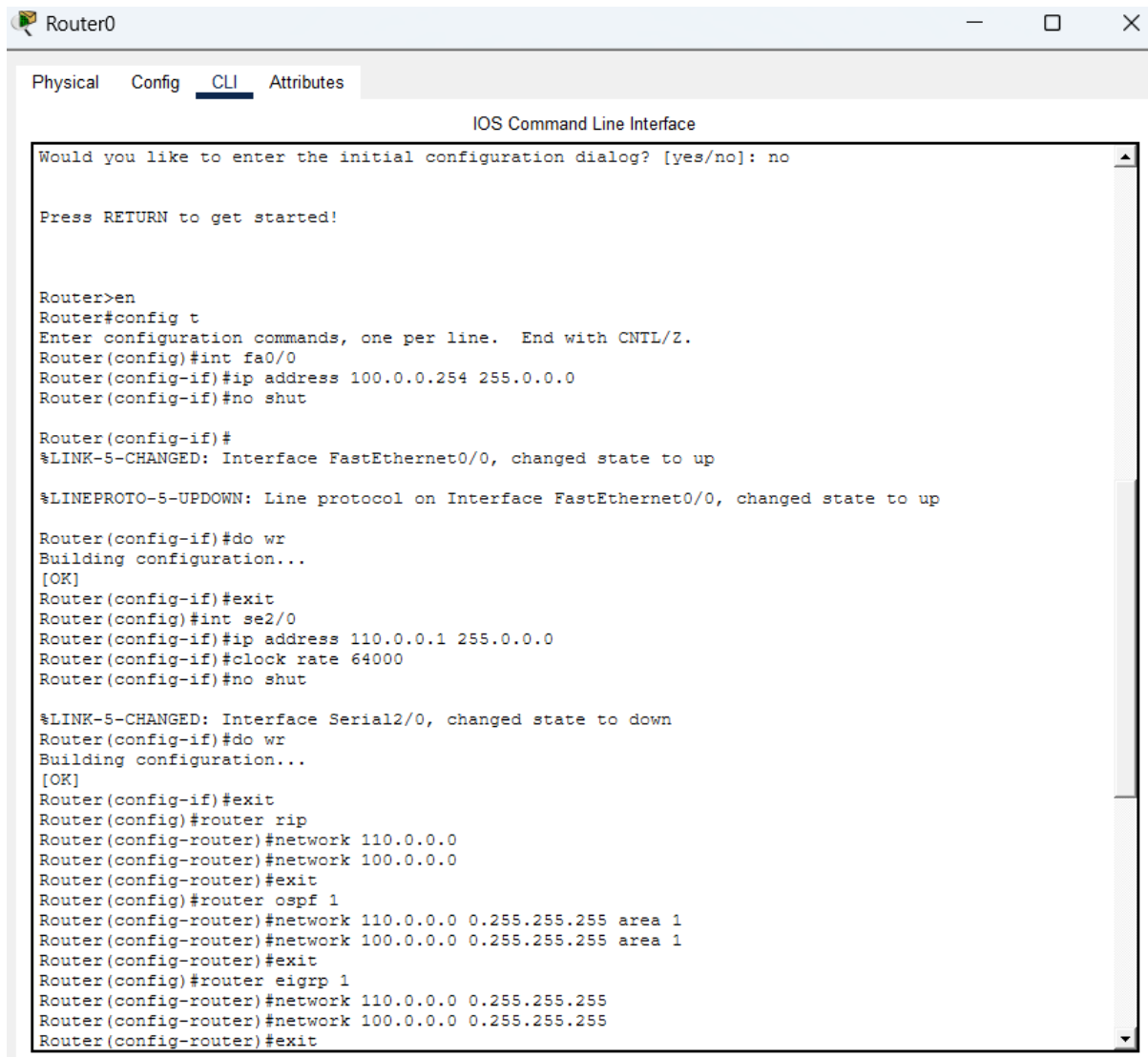


Figure 2: Router0 configuration with OSPF, RIP and EIGRP

For Router 1:

se2/0

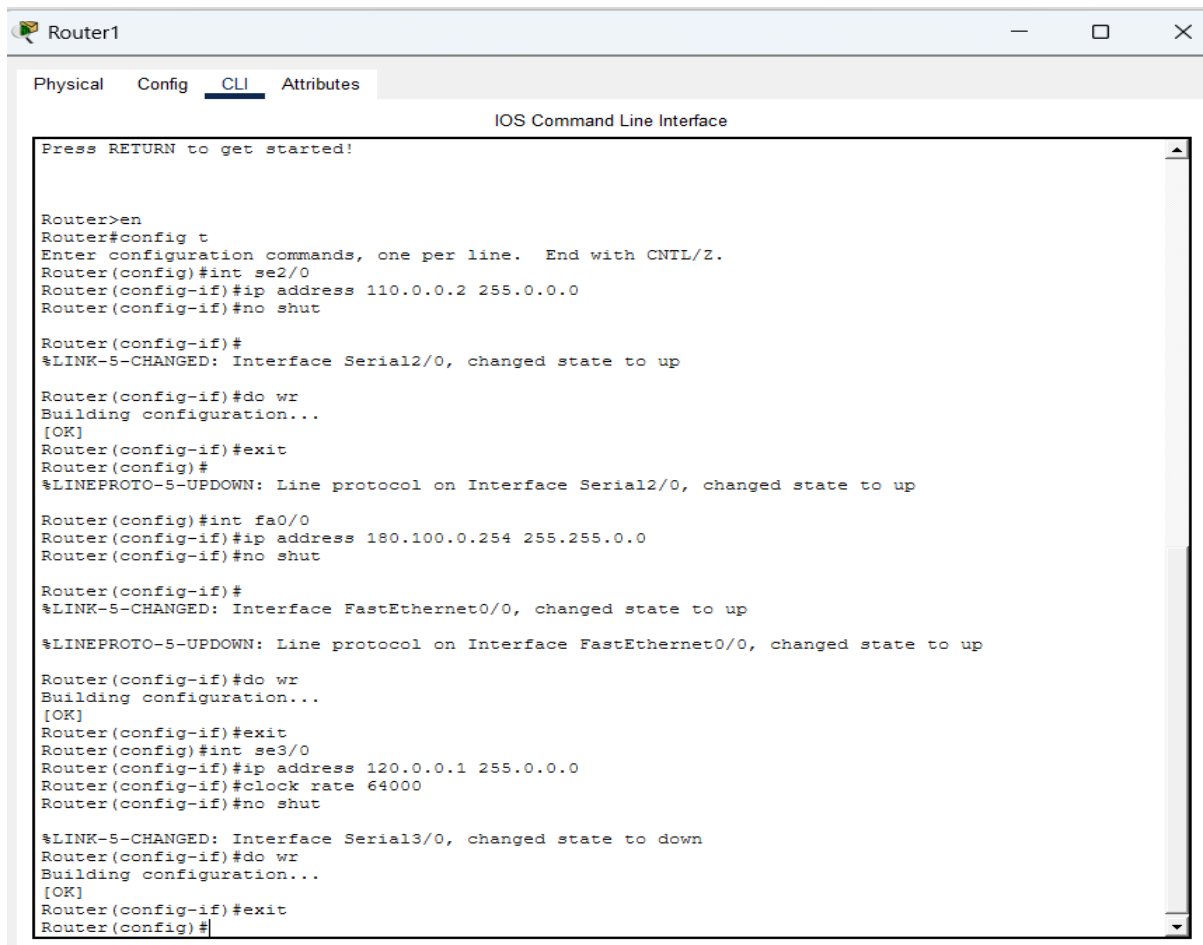
```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 110.0.0.2 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 180.100.0.254 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

se3/0

```
Router(config)#int se3/0
Router(config-if)#ip address 120.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```



```
Router1
Physical Config CLI Attributes
IOS Command Line Interface
Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 110.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#int fa0/0
Router(config-if)#ip address 180.100.0.254 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int se3/0
Router(config-if)#ip address 120.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Router1 (RIP Configuration)

```
Router(config)#router rip
Router(config-router)#network 110.0.0.0
Router(config-router)#network 180.100.0.0
Router(config-router)#network 120.0.0.0
Router(config-router)#exit
```

```
Router(config)#router rip
Router(config-router)#network 110.0.0.0
Router(config-router)#network 180.100.0.0
Router(config-router)#network 120.0.0.0
Router(config-router)#exit
```

Router1 (OSPF Configuration)

```
Router(config)#router ospf 1
Router(config-router)#network 110.0.0.0 0.255.255.255 area 1
Router(config-router)#network 180.100.0.0 0.0.255.255 area 1
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#exit
```

```
Router(config)#router ospf 1
Router(config-router)#network 110.0.0.0 0.255.255.255 area 1
Router(config-router)#network 180.100.0.0 0.0.255.255 area 1
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
```

Router1 (EIGRP Configuration)

```
Router(config)#router eigrp 1
Router(config-router)#network 110.0.0.0 0.255.255.255
Router(config-router)#network 180.0.0.0 0.0.255.255
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#exit
```

```
Router(config)#router eigrp 1
Router(config-router)#network 110.0.0.0 0.255.255.255
Router(config-router)#network 180.0.0.0 0.0.255.255
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#exit
```

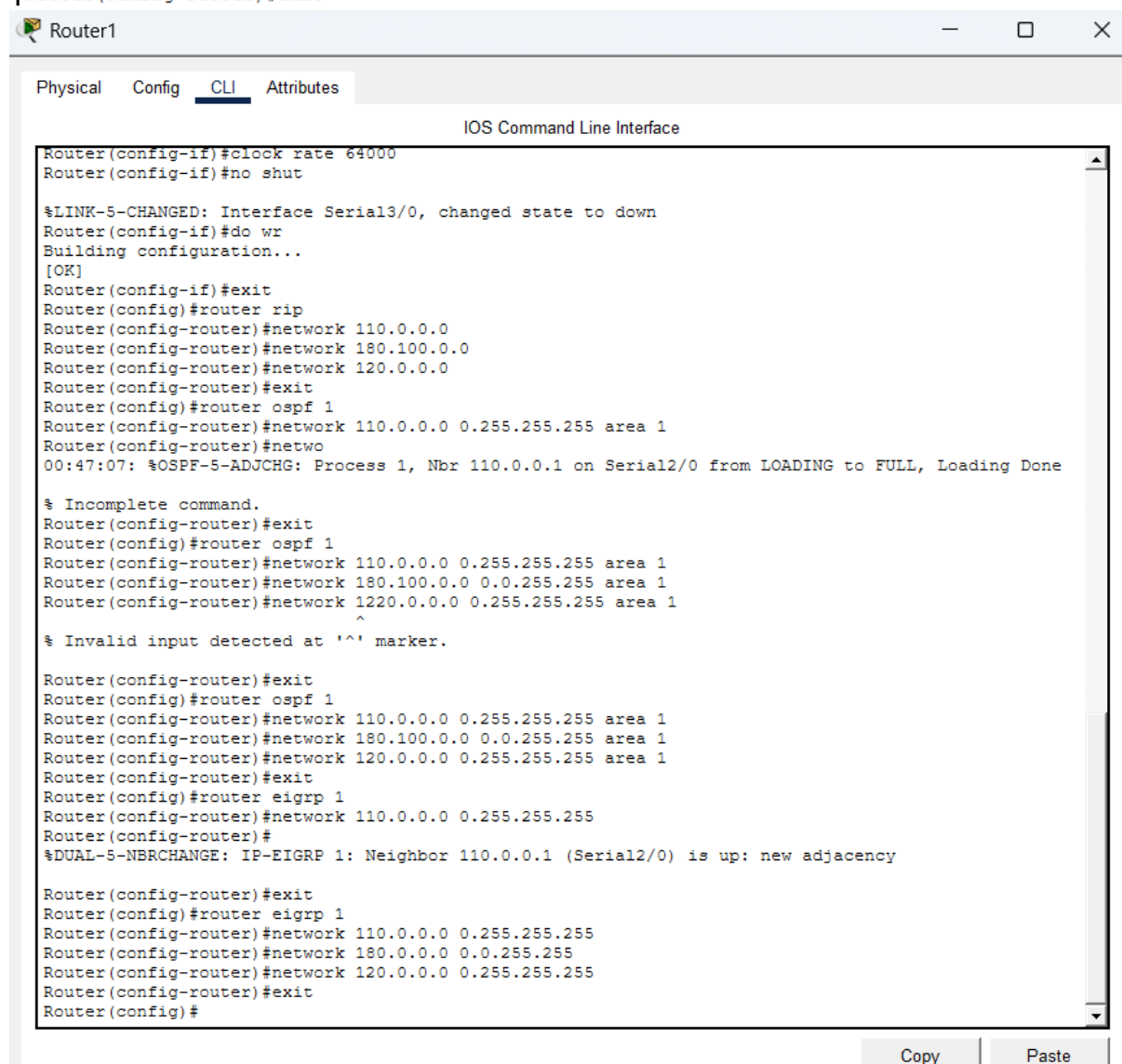


Figure 3: Router1 configuration with OSPF, RIP and EIGRP

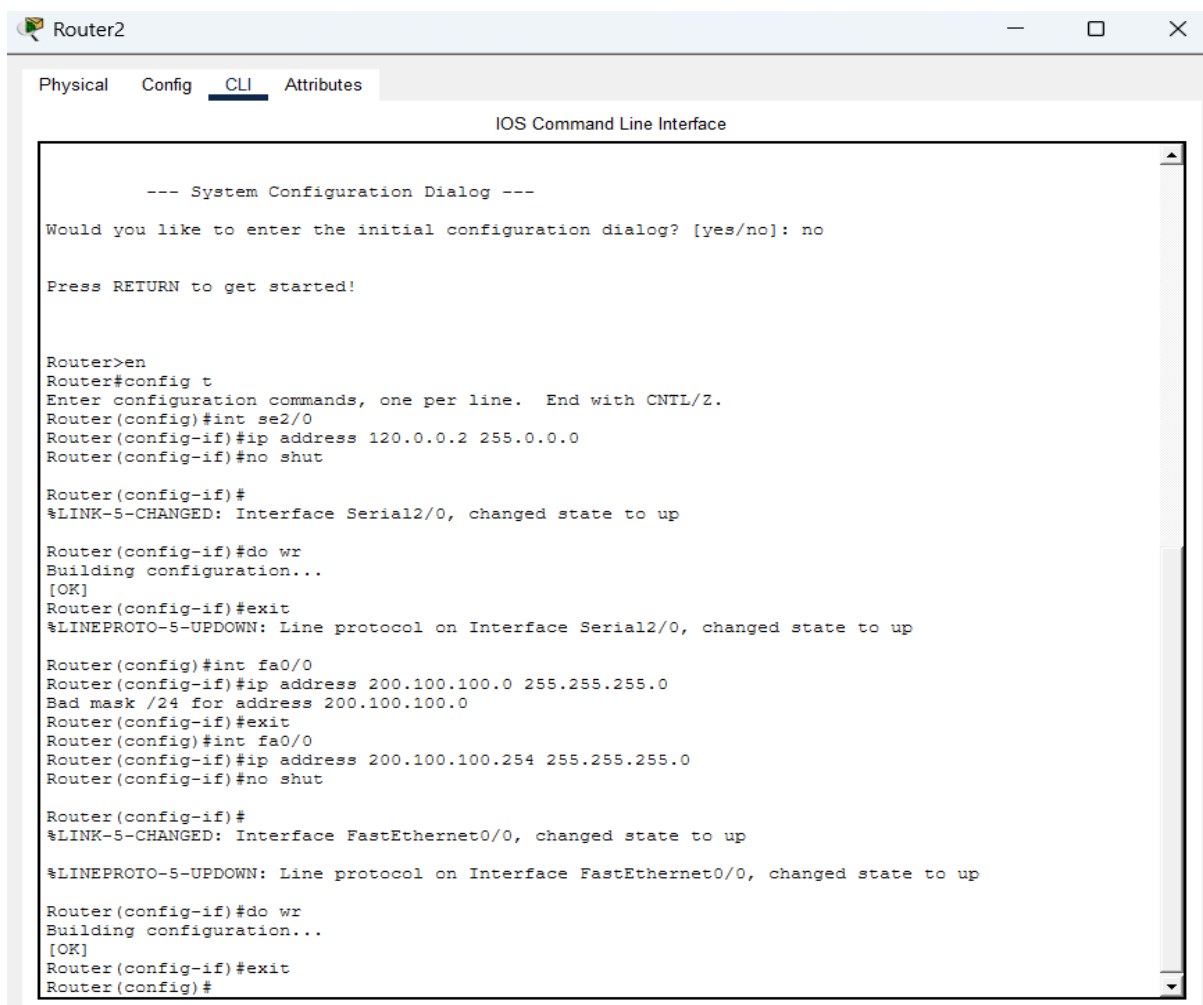
For Router 2:

se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 120.0.0.2 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 200.100.100.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```



The screenshot shows a window titled "Router2" with a tabbed interface. The "CLI" tab is selected, displaying the "IOS Command Line Interface". The terminal output shows the following sequence of commands and responses:

```
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 120.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#int fa0/0
Router(config-if)#ip address 200.100.100.0 255.255.255.0
Bad mask /24 for address 200.100.100.0
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 200.100.100.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#
```

Router2 (RIP Configuration)

```
Router(config)#router rip
```

```
Router(config-router)#network 120.0.0.0
Router(config-router)#network 200.100.100.0
Router(config-router)#exit
```

```
Router(config)#router rip
Router(config-router)#network 120.0.0.0
Router(config-router)#network 200.100.100.0
Router(config-router)#exit
```

Router2 (OSPF Configuration)

```
Router(config)# router ospf 1
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#network 200.100.100.0 0.0.0.255 area 1
Router(config-router)#exit
```

```
Router(config)# router ospf 1
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#
01:09:43: %OSPF-5-ADJCHG: Process 1, Nbr 180.100.0.254 on Serial2/0 from LOADING to FULL, Loading Done
Router(config-router)#network 200.100.100.0 0.0.0.255 area 1
Router(config-router)#exit
```

Router2 (EIGRP Configuration)

```
Router(config)# router eigrp 1
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#network 200.100.100.0 0.0.0.255
Router(config-router)#exit
```

```
Router(config)# router eigrp 1
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#
%DUAL-S-NBRCHANGE: IP-EIGRP 1: Neighbor 120.0.0.1 (Serial2/0) is up: new adjacency
Router(config-router)#network 200.100.100.0 0.0.0.255
Router(config-router)#exit
```

The screenshot shows a window titled "Router2" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following sequence of commands and messages:

```
$LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#int fa0/0
Router(config-if)#ip address 200.100.100.0 255.255.255.0
Bad mask /24 for address 200.100.100.0
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 200.100.100.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 120.0.0.0
Router(config-router)#network 200.100.100.0
Router(config-router)#exit
Router(config)# router ospf 1
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#
01:09:43: %OSPF-5-ADJCHG: Process 1, Nbr 180.100.0.254 on Serial2/0 from LOADING to FULL, Loading Done

Router(config-router)#network 200.100.100.0 0.0.0.255 area 1
Router(config-router)#exit
Router(config)# router eigrp 1
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 120.0.0.1 (Serial2/0) is up: new adjacency

Router(config-router)#network 200.100.100.0 0.0.0.255
Router(config-router)#exit
Router(config)#
```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons.

Figure 4: Router2 configuration with OSPF, RIP and EIGRP

Ping Communication:

The screenshot shows a Cisco Packet Tracer PC window with the 'Desktop' tab selected. Inside the window is a 'Command Prompt' application. The command prompt displays the output of three ping commands: `ping 180.100.0.1`, `ping 180.100.0.2`, and `ping 200.100.100.1`. Each command shows four successful replies with varying round-trip times and TTL values, followed by a summary of the ping statistics. The window title bar indicates it is a PC1, and the tabs at the top are Physical, Config, Desktop, Programming, and Attributes.

```
PC1
Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 180.100.0.1

Pinging 180.100.0.1 with 32 bytes of data:

Reply from 180.100.0.1: bytes=32 time=24ms TTL=126
Reply from 180.100.0.1: bytes=32 time=12ms TTL=126
Reply from 180.100.0.1: bytes=32 time=11ms TTL=126
Reply from 180.100.0.1: bytes=32 time=2ms TTL=126

Ping statistics for 180.100.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 24ms, Average = 12ms

C:\>ping 180.100.0.2

Pinging 180.100.0.2 with 32 bytes of data:

Reply from 180.100.0.2: bytes=32 time=17ms TTL=126
Reply from 180.100.0.2: bytes=32 time=10ms TTL=126
Reply from 180.100.0.2: bytes=32 time=2ms TTL=126
Reply from 180.100.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 180.100.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 17ms, Average = 7ms

C:\>ping 200.100.100.1

Pinging 200.100.100.1 with 32 bytes of data:

Reply from 200.100.100.1: bytes=32 time=10ms TTL=125
Reply from 200.100.100.1: bytes=32 time=3ms TTL=125
Reply from 200.100.100.1: bytes=32 time=11ms TTL=125
Reply from 200.100.100.1: bytes=32 time=3ms TTL=125

Ping statistics for 200.100.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 11ms, Average = 6ms

C:\>ping 200.100.100.2

Pinging 200.100.100.2 with 32 bytes of data:
```

Figure 5: Ping Command

```

PC1
Physical Config Desktop Programming Attributes
Command Prompt
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 2ms, Maximum = 24ms, Average = 12ms

C:\>ping 180.100.0.2

Pinging 180.100.0.2 with 32 bytes of data:

Reply from 180.100.0.2: bytes=32 time=17ms TTL=126
Reply from 180.100.0.2: bytes=32 time=10ms TTL=126
Reply from 180.100.0.2: bytes=32 time=2ms TTL=126
Reply from 180.100.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 180.100.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 17ms, Average = 7ms

C:\>ping 200.100.100.1

Pinging 200.100.100.1 with 32 bytes of data:

Reply from 200.100.100.1: bytes=32 time=10ms TTL=125
Reply from 200.100.100.1: bytes=32 time=3ms TTL=125
Reply from 200.100.100.1: bytes=32 time=11ms TTL=125
Reply from 200.100.100.1: bytes=32 time=3ms TTL=125

Ping statistics for 200.100.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 11ms, Average = 6ms

C:\>ping 200.100.100.2

Pinging 200.100.100.2 with 32 bytes of data:

Reply from 200.100.100.2: bytes=32 time=2ms TTL=125
Reply from 200.100.100.2: bytes=32 time=10ms TTL=125
Reply from 200.100.100.2: bytes=32 time=11ms TTL=125
Reply from 200.100.100.2: bytes=32 time=11ms TTL=125

Ping statistics for 200.100.100.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 11ms, Average = 8ms

C:\>

```

Figure 6: Ping Command

Ping Test Results Between PCs

Realtime Sim										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC5	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	PC3	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC2	PC5	ICMP		0.000	N	3	(edit)	(delete)